

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer **ALL** questions in this section. Marks are indicated at the end of each question. Together they are worth 80% of the total marks for this examination.

Question 1 (18 marks – approximately 32 minutes)

Selecting an appropriate implementation strategy is an important process when an organisation is going to replace its old and legacy accounting system with a new one.

Required:

- (a) Contrast the difference between a "direct cutover" implementation strategy and a "side-by-side" implementation strategy.

(4 marks)

- (b) Justify why a "side-by-side" implementation strategy is the least risky approach in system conversion.

(2 marks)

As accountants are the major users of computerised accounting systems, they are expected to participate actively in some or even all stages of system conversion.

Required:

- (c) State any TWO duties of accountants in each of the following stages during a system conversion.

- (i) testing; and

(3 marks)

- (ii) operation and maintenance.

(3 marks)

A successful system change usually requires some techniques for both addressing user resistance and motivating user adoption.

Required:

- (d) Demonstrate any TWO ways that an organisation can support a system change through revising management policies.

(6 marks)

Question 2 (18 marks – approximately 32 minutes)

- (a) A new type of fingerprint security integrated circuit has been announced and introduced by a Korean semiconductor technology company. Banks may issue biometric payment cards with this technology to enhance authentication security. Such technology can also be used for student or employee identification, membership, or access to buildings.

Required:

- (i) Define "biometric authentication".
(2 marks)
- (ii) Demonstrate any THREE biometric factors of using a fingerprint authentication system.
(6 marks)

Man-in-the-middle (MITM) attacks are a common type of cyberattack. Hackers usually use MITM attacks to steal login credentials or personal information, or even to spy on individuals or groups.

Required:

- (b) Demonstrate how hackers could conduct a MITM attack through.
- (i) a wireless router or a wireless access point; and
(3 marks)
- (ii) a web browser.
(3 marks)
- (c) (i) Define virtual private network (VPN).
(2 marks)
- (ii) Explain why a VPN could be used to prevent the MITM attack as stated in part (b)(i) above.
(2 marks)

Question 3 (18 marks – approximately 32 minutes)

(a) Categorise the following information into strategic level, tactical level or operational level:

- (i) findings from a SWOT analysis;
- (ii) a manpower utilisation report;
- (iii) a list which shows real-time item availability among all the inventories;
- (iv) daily production schedules;
- (v) a sales forecast for the coming two years; and
- (vi) a monthly departmental salary and wages report.

(6 marks)

(b) Describe the basic function per each of the following types of information systems:

- (i) a transaction processing system (TPS);
- (ii) a management information system (MIS);
- (iii) an expert system (ES); and
- (iv) an executive information system (EIS).

(6 marks)

(c) State and describe any THREE types of ethical or legal issues that organisations may face when using information systems.

(6 marks)

Question 4 (18 marks – approximately 32 minutes)

(a) List any TWO types of information needs per each of the following function areas:

(i) human resources; and

(2 marks)

(ii) marketing.

(2 marks)

(b) Although cloud computing is becoming popular nowadays, many organisations are still running their information systems with a client-server model on their premises.

Required:

(i) Contrast any THREE differences between a server and a client.

(6 marks)

(ii) Justify the importance of using a network switch in a client-server network. You should start by describing the functions of a network switch.

(3 marks)

Question 5 (13 marks – approximately 24 minutes)

(a) Determine any TWO risks that may occur if an e-commerce business has weak identity authentication among its customers and suppliers.

(4 marks)

(b) Contrast the differences between e-commerce and e-business.

(5 marks)

(c) According to a global survey conducted by an international CPA firm in 2021, 78% of respondents are currently implementing Robotic Process Automation (RPA) and 16% of respondents plan to do so in the next three years.

Required:

(i) Define "Robotic Process Automation" (RPA).

(2 marks)

(ii) List any TWO examples of using RPA in accounting.

(2 marks)

* * * END OF EXAMINATION PAPER * * *